FINTECHATHON 2026 · TOPIC E · PROJECT PROPOSAL

Setu (সেতু)

An AI finance copilot that helps Bangladeshi SMEs finance, plan and safely pay their Chinese suppliers.

The AI explains. Code decides.

Simulated sandbox · no real money

No model training

Bangladesh ⇄ China trade finance · Milestone escrow · Document forensics · Risk-tiered human approval

THE PROBLEM

Importing from China is a cash-flow and trust gamble for SMEs

1 · Unseen cash gaps

Orders of USD 8,000–35,000 are paid in tranches by LC or TT. The real gap (goods + freight + duty + VAT + bank charges + FX − cash on hand) is rarely computed before a payment is due, so production stalls.

2 · Document fraud

A forged bill of lading, a doctored invoice, or the same invoice financed at two lenders can cost the entire advance. Hidden text in PDFs can even instruct an AI assistant to "release payment".

3 · Payment redirection

Supplier bank details change by email. Licence name, invoice name and receiving-account name quietly stop matching, and personal or newly opened accounts receive the money.

Gap in today's tools: generic chatbots can explain but cannot be trusted with money; banks' systems can move money but do not reason across the whole order.

WHO WE SERVE

One persona, one order, followed end to end

Rahima — Rahman Textiles, Dhaka

- Fabric importer, 12 staff, buys from Guangdong suppliers
- Home currency BDT; pays in USD or CNY
- Order SETU-2026-0001: 8,000 m fabric, goods USD 9,600
- Landed cost USD 11,200 vs cash USD 6,000
- Three legitimate suppliers and one suspicious one

What Setu answers for her

- "How much am I short, and when?" — **USD 1,560 on 7 Nov 2026**
- "Which financing is cheapest and allowed?"
- "Is this supplier and this document genuine?"
- "Can I release the next tranche safely?"
- Asked in English, Bangla, or (supplier view) Chinese

3
financing options compared on total BDT cost, rate, tenor and eligibility

90
day cash-flow calendar with T-7 / T-3 / T-1 deadline alerts

3
languages: English, Bangla, Chinese

PROPOSED SOLUTION

Nine capabilities in one copilot

Financing match

Computes the financing gap and ranks supplier credit, bank import loan and escrow-backed loan with reasons.

Cash-flow alerts

Day-by-day 90-day calendar; shortfall, deadline and FX-timing alerts.

Landed cost & margin

Duty, VAT, AIT by HS code, freight, insurance, bank and FX costs; supplier scorecard.

Supplier KYB

Fuzzy match of licence, invoice and account names; flags personal and new accounts.

Milestone escrow

30/60/10 schedules on a simulated double-entry ledger; tranches release only after evidence checks pass.

Document forensics

Cross-checks invoice, packing list, B/L and certificate of origin; detects hidden text.

USD vs CNY route

Compares total BDT cost via each currency; mock rate lock for N hours.

Signed credit record

Ed25519-signed record of each completed deal that a lender can verify.

Explained chat

Plain-language answers in EN / BN / ZH that cite the numbers and rules used.

HOW IT WORKS

From question to safe payment

1. **Ask** — Rahima asks in Bangla or English
2. **Analyse** — Landed cost, gap and cash-flow computed in code
3. **Finance** — Options ranked; shortfall covered
4. **Verify** — KYB and document forensics; duplicate hash check
5. **Approve** — Risk tier decides: auto, one-tap, or OTP + cooling-off
6. **Release** — Escrow tranche paid (simulated), audit-logged
7. **Record** — Signed credit record on completion

Two "wow" moments in the live demo

- **Forged B/L:** hidden white text is exposed, check FAILS, escalates to Tier 3, chat states it ignored the instruction.
- **Duplicate financing:** the same invoice offered to a second lender is blocked by the shared hash registry.

Always honest about "simulated"

Every money action is labelled **Simulated**. Lenders are fictional and FX spreads are mock. No real payment rails are connected.

SECURITY BY DESIGN

Permission tiers enforced in code, not in prompts

Tier	Behaviour	Triggers
Tier 0	Automatic	Analysis and checks (read-only)
Tier 1	Automatic + notice	Release to a verified, unchanged beneficiary; all checks pass; ≤ USD 20,000
Tier 2	One-tap human approval	First deal with a supplier, minor mismatch, FX lock, milestone schedule creation or change
Tier 3	OTP + 10-minute cooling-off	Beneficiary change, high fraud score, duplicate hash, amount > USD 20,000, **any instruction found in document text**

LLM can only propose

Tools exposed to the model are Tier 0/1 only. Tier 2/3 create a pending approval that only a human HTTP endpoint can execute.

Injection & privacy

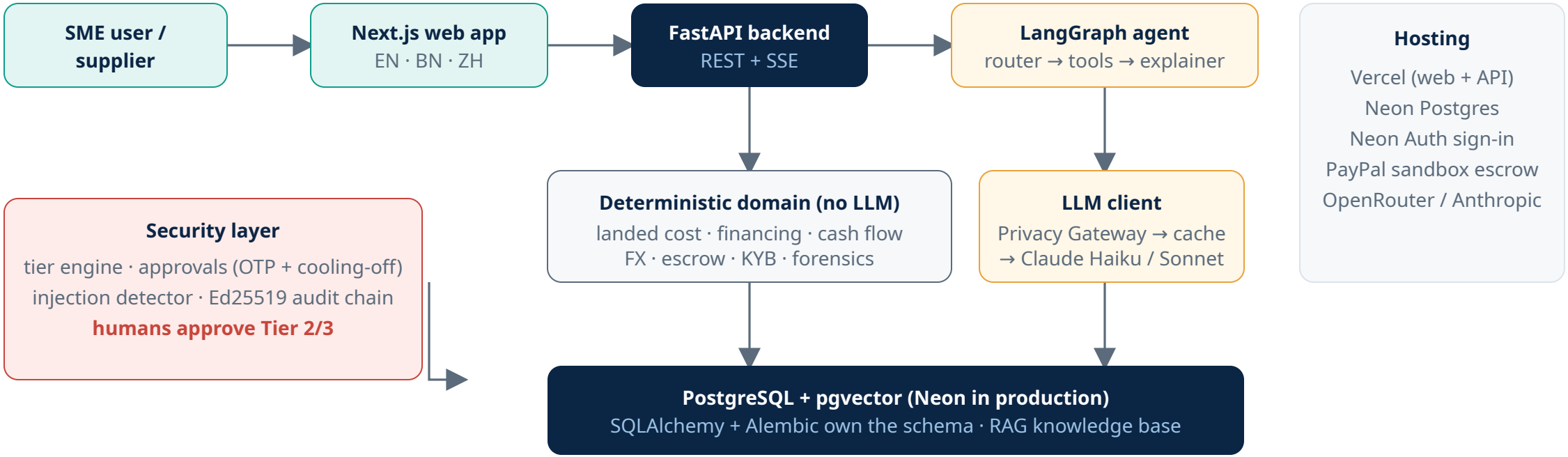
Document text is untrusted data. A Privacy Gateway masks NID, accounts, phones and prices with reversible tokens before any model call.

Tamper-evident audit

Every action is in an Ed25519-signed, hash-chained log, verifiable by anyone with the public key.

ARCHITECTURE

A thin AI layer over deterministic finance code



AI APPROACH

"LLM explains, code executes"

What the model does

- Reads tool outputs and explains them in plain EN / BN / ZH
- Routes questions to the right Tier 0/1 tool
- Retrieves knowledge-base passages (pgvector + full-text hybrid RAG)
- Haiku-class model for chat; Sonnet-class only for document forensics

What the model never does

- Compute any number: cost, margin, gap, FX, fraud score or tier
- Execute Tier 2/3 actions or move money
- Treat document or email text as instructions
- See raw identifiers: they are tokenised first

Engineering guarantees

- Money is Decimal, never float
- One module is the only place models are called
- Identical masked inputs are served from cache (cost control)
- No model training; only pre-trained APIs are used

EVIDENCE

Measured results (development split, synthetic data)

1.00

forensics precision and recall across 11 checks and 10 fraud types; 0 of 50 clean packs flagged

5/5

duplicate-financing attempts caught (3 exact, 2 re-saved)

100%→0%

prompt-injection attack success, defences off → on (13 attacks); 0/300 false positives; no money moved

0

detectable identifiers reach the model with masking on (2,490 with it off)

Read these honestly

Data is synthetic: 100 clean and 100 tampered document packs, five injection styles. Perfect dev scores are a ceiling, not real-world accuracy, because the generator and the checks share a team. The held-out test split will be run once and reported as is.

INNOVATION

What makes Setu different

Explained risk tiers

Every approval shows a rule trace and "what would change it" hints, so users learn why a payment was escalated.

Privacy X-ray

Users can inspect exactly what the model saw, masked tokens included.

Live attack console

Upload a document with a hidden instruction and watch the defences on vs off.

Publicly verifiable trust

Anyone can verify the audit log or a signed credit record in the browser with a public key, no login.

Escrow plus credit history

Completed, on-time deals become a signed record that can unlock better financing for thin-file SMEs.

Built for Bangladesh

Bangla-first chat, BDT landed-cost rules by HS code, and a CNY route that avoids the double USD conversion.

DELIVERY PLAN

Phased build, working version by 19 Oct 2026

Stage	Scope	Status
1 · Foundation	Scaffold, schema, seed data, synthetic trade-document generator	**Done**
2 · Finance engines	Landed cost, financing match, cash flow, FX route, dashboard UI	**Done**
3 · Agent & RAG	LLM plumbing, hybrid RAG, LangGraph agent, multilingual chat	**Done**
4 · Trust	Escrow, KYB, forensics, hash registry, signed credit record	**Done**
5 · Security	Tiers, approvals with OTP, injection defence, privacy gateway, audit log	**Done**
6–7 · Demo & evaluation	Guided demo, landing page, datasets, evaluation harness	**Done (dev split)**
8–9 · Deploy & submit	Vercel + Neon deployment, documentation, security self-assessment, demo video	**Done / finalising**
Next · Production path	Neon Auth onboarding with KYB gate, PayPal sandbox escrow, step-up auth, test-split report	**In progress**

RISKS & ROADMAP

Known limits and how we address them

Limits today

- Synthetic data; dev-split numbers only until the test split runs
- Lenders are fictional and FX spreads are mock
- Sandbox money only; no real payment rails
- Bangla and Chinese not yet reviewed by native speakers
- Photo-versus-spec check is colour only

Roadmap

- Report the held-out test split once per configuration
- Real authentication and SMS delivery for OTP
- Lender and payment-sandbox integration (PayPal sandbox first)
- Vision-model photo checks; OCR for dimensions
- Native-speaker review; pilot with real importers and a partner bank

Compliance posture

Sandbox-only by design; secrets live in environment variables and never reach the model; no third-party template code or assets are used.

IN SUMMARY

Setu is the bridge between a Bangladeshi importer and a Chinese supplier: fewer cash surprises, fewer fraud losses, and payments a human can always trust.

Topic E coverage: financing · cash flow · analysis · KYB · escrow · forensics · FX · credit record · chat

Security: tiers in code · injection defence · privacy gateway · signed audit log

Thank you · Questions welcome